

EECE4632 Hardware-Software Codesign for FPGA Based Systems

Project Update 2

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Abstract

For our Final Project, we are going to design a Guitar Pedal with audio processing algorithms on the AUP-ZU3 board. We will use the Audio Codec on the board to input the sound, and design specified algorithms to control the sound in C. Then we will create overlays to run the algorithms on the hardware, and compare it to the speed of the algorithms in Python software. We will output the modified sound through the board's DAC to a speaker, and hear the changes in real time.

What we implemented for Project Update 1

We completed the algorithms for hard and soft clipping distortion, tremolo, echo, and chorus in Python for our software implementation. We created a Vitis .cpp file to test our wav file input and output on the AUP-ZU3 audio codec.

Since Project Update 1

We focused a LOT on testing the input wav file and output audio codec on the board, creating an overlay from the Vitis code and altering the Vivado design many times. We changed our design to utilize AXI Stream instead of AXI Lite and created custom outputs for the board in the Vivado design.

After we still could not get this to work, David kept focusing on debugging the Vivado, finding issues with the I2C drivers and I2S connections. Clara started writing the C++ code for the effects, translating them from our existing Python code. She got the hard and soft clipping distortion working as well as the echo, verifying them with a testbench with a simulated sine wave.

Right now we have no code to run because the overlay is still not officially working with the input and output.

Plan for the remainder of the semester

This next week:

- 1) We will get the input and output working with the overlay

- 2) We will implement the tremolo and chorus effects in C++, as well as a toplevel to run them all

The week after that:

- 1) We will alter the DATA_SIZE array inputs to our functions so we can concatenate the different effects
- 2) We will test all the effects and clean up anything that is necessary. This could include making a pretty graphic/window for the user to select different effects.
- 3) We will compile data and examples and create our project presentation for our presentation on April 15th.